

IOT ASSIGNMENT

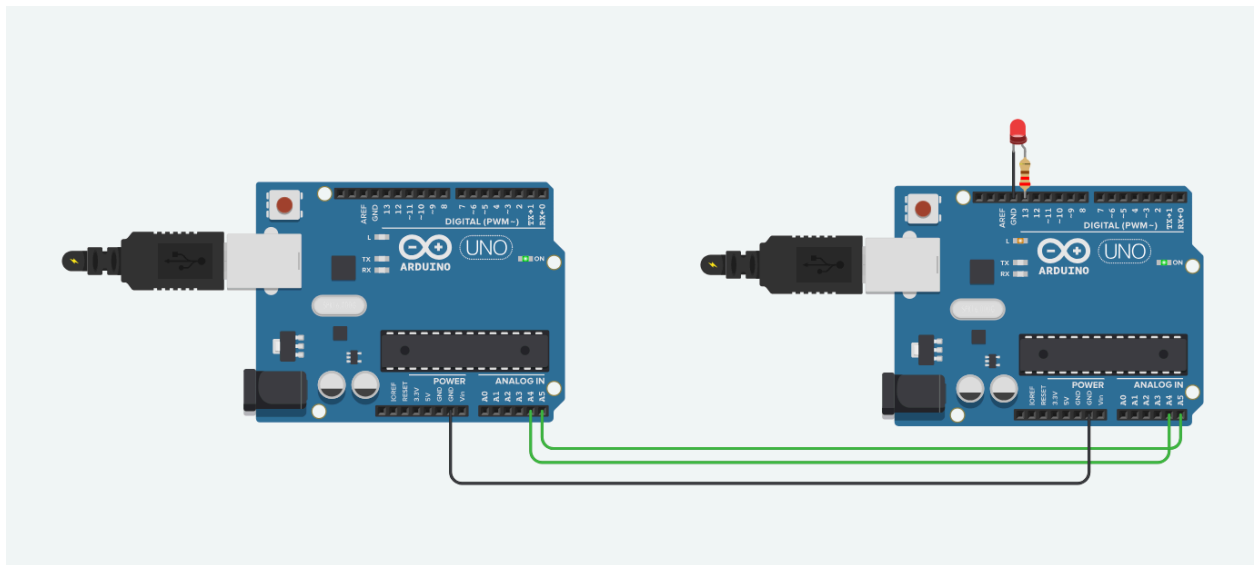
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1) How to enable sleep mode in ESP32?

ESP32 supports multiple power-saving modes including Modem Sleep, Light Sleep, Deep Sleep, and Hibernation. Deep Sleep is commonly used in IoT applications. Before entering sleep, wake-up sources such as timers, GPIO interrupts, or touch sensors are configured using ESP-IDF or Arduino functions. The ESP32 then enters low-power mode and returns to executing.

To wake-up a source such as a timer, GPIO pin, or touch sensor, we use functions like **esp_sleep_enable_timer_wakeup()** or **esp_sleep_enable_ext0_wakeup()**. After configuring the wake-up source, **esp_deep_sleep_start()** is called to enter Deep Sleep mode. When the wake-up condition occurs, the ESP32 restarts execution from the **setup()** function.

2) Setting up I2C in tinkercad.



Master code:

```
#include <Wire.h>
```

```
void setup(){
```

```
  Wire.begin();}
```

```
void loop(){
```

```
  Wire.beginTransmission(8);
```

```
  Wire.write(1);
```

```
  Wire.endTransmission();
```

```
delay(1000);
Wire.beginTransmission(8);
Wire.write(0);
Wire.endTransmission();
delay(1000);}
```

Slave code:

```
#include <Wire.h>
int led = 13;
```

```
void setup(){
  pinMode(led, OUTPUT);
  Wire.begin(8);
  Wire.onReceive(receive);}
```

```
void loop(){} 
```

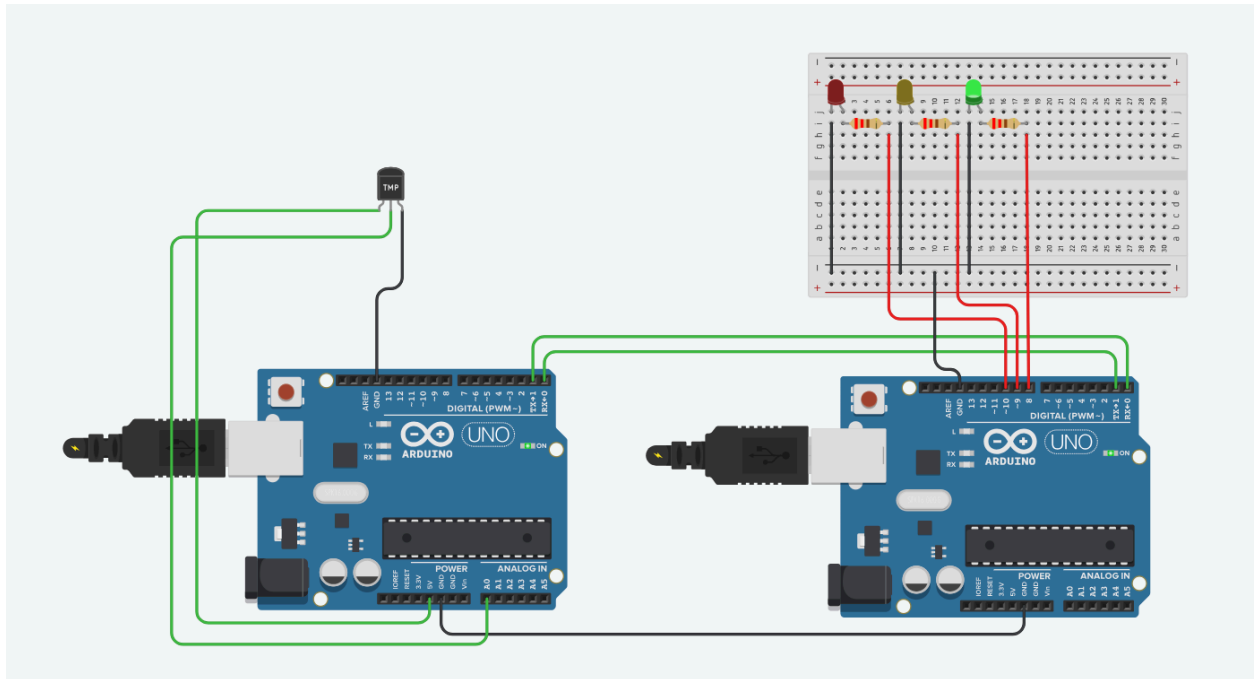
```
void receive(int bytes){
  int x = Wire.read();
  if(x == 1){
    digitalWrite(led, HIGH);}
  else{
    digitalWrite(led, LOW);}}
```

3) Write a code to set up SPI.

```
#include <SPI.h>
void setup(){
  SPI.begin();
  pinMode(10, OUTPUT);
  Serial.begin(9600);
  Serial.println("SPI Started");}
```

```
void loop(){
  digitalWrite(10, LOW);
  SPI.transfer('A');
  digitalWrite(10, HIGH);
  delay(1000);}
```

4) Establish Master Slave tx/rx between 2 arduino. Uart. Try transferring data.



Master code:

```
int sensor = A0;
void setup(){
  Serial.begin(9600);}

void loop(){
  int reading = analogRead(sensor);
  float voltage = reading * (5.0 / 1023.0);
  float temp = (voltage - 0.5) * 100;
  Serial.println(temp);
  delay(1000);}
```

Slave code:

```
float temp;

void setup(){
  Serial.begin(9600);
  pinMode(8, OUTPUT);
  pinMode(9, OUTPUT);
  pinMode(10, OUTPUT);}

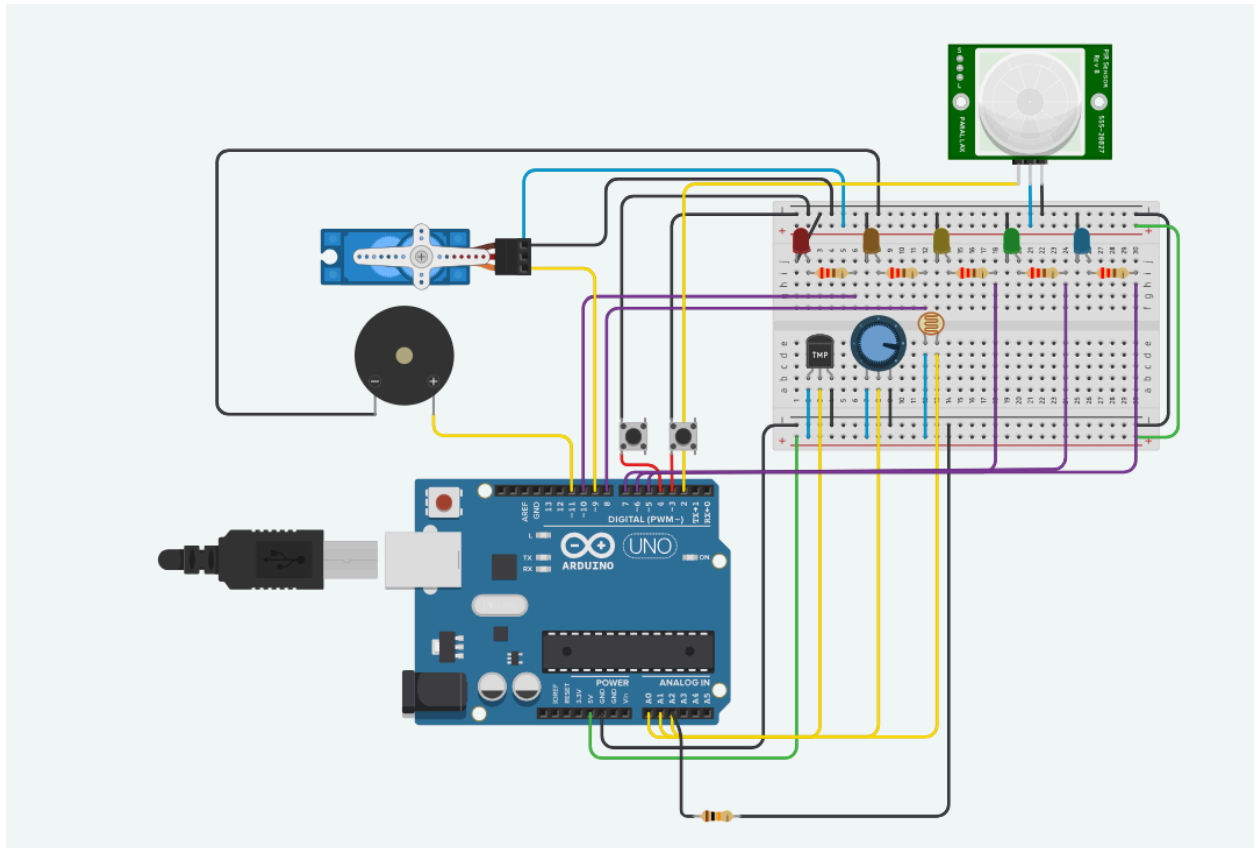
void loop(){
  if(Serial.available()){
```

```

temp = Serial.parseFloat();
if(temp < 25){
digitalWrite(8, HIGH);
digitalWrite(9, LOW);
digitalWrite(10, LOW);}
else if(temp < 35){
digitalWrite(8, LOW);
digitalWrite(9, HIGH);
digitalWrite(10, LOW);}
else{
digitalWrite(8, LOW);
digitalWrite(9, LOW);
digitalWrite(10, HIGH);}}}

```

5) Utilize all pins of Arduino UNO.



Code:

```
#include <Servo.h>
Servo door;
int tempsensor = A0;
int fuelsensor = A1;
int ldr = A2;

int irsensor = 2;
int lock = 3;
int unlock = 4;

int bled = 5; // cold
int gled = 6; // warm
int yled = 7; // hot

int oled = 8; // headlight
int rled = 10; // fuel
int buzzer = 11;

void setup(){
  pinMode(bled, OUTPUT);
  pinMode(gled, OUTPUT);
  pinMode(yled, OUTPUT);
  pinMode(oled, OUTPUT);
  pinMode(rled, OUTPUT);
  pinMode(buzzer, OUTPUT);

  pinMode(irsensor, INPUT);
  pinMode(lock, INPUT_PULLUP);
  pinMode(unlock, INPUT_PULLUP);

  digitalWrite(lock, HIGH);
  digitalWrite(unlock, HIGH);

  door.attach(9);
  door.write(0);
  Serial.begin(9600);}

void loop(){
  int reading = analogRead(tempsensor);
  float voltage = reading * (5.0 / 1023.0);
  float temp = (voltage - 0.5) * 100;

  if(temp < 25){
```

```

    digitalWrite(bled, HIGH);
    digitalWrite(gled, LOW);
    digitalWrite(yled, LOW);}
else if(temp < 35){
    digitalWrite(bled, LOW);
    digitalWrite(gled, HIGH);
    digitalWrite(yled, LOW);}
else{
    digitalWrite(bled, LOW);
    digitalWrite(gled, LOW);
    digitalWrite(yled, HIGH);}

int lightLevel = analogRead(ldr);
if(lightLevel < 500){
    digitalWrite(oled, HIGH);}
else{
    digitalWrite(oled, LOW);}

int fuel = analogRead(fuelsensor);
if(fuel < 300){
    digitalWrite(rled, HIGH);}
else{
    digitalWrite(rled, LOW);}

if(digitalRead(irsensor) == HIGH){
    tone(buzzer, 1000);}
else{
    noTone(buzzer);}

if(digitalRead(lock) == LOW){
    door.write(0);
    Serial.println("Door Locked");
    delay(300);}

if(digitalRead(unlock) == LOW){
    door.write(90);
    Serial.println("Door Unlocked");
    delay(300);}

Serial.print("Temp: ");
Serial.print(temp);
Serial.print(" Fuel: ");
Serial.print(fuel);
Serial.print(" Light: ");

```

```
Serial.println(lightLevel);  
delay(200);}
```